7. Cross-checking with new database

To now test the robustness of our classification model, we tried to apply our metrics on a new database :

CGMacros v1.0.0 — PhysioNet · doi:10.13026/3z8q-x658 · Gutiérrez-Osuna, Kerr, Mortazavi y Das (2025) · licencia CC BY-NC-SA 4.0

Producto: 45 archivos .mat repartidos en MATNDBT2, MATPDBT2 y MATDBT2

It is also important to notice that this time the technicians used two probes to measure the glucose concentration, called `dexcom` and `libre`. We can then double the process of verification.

Important : This time, the sampling frequency is one point every minute so we need to do a quick downsample in order to compare every data on a 15-minute point basis

Let's see first how the points behave on the HxC feature plane. For now he still use **($m = 5, \tau = 2$)** :

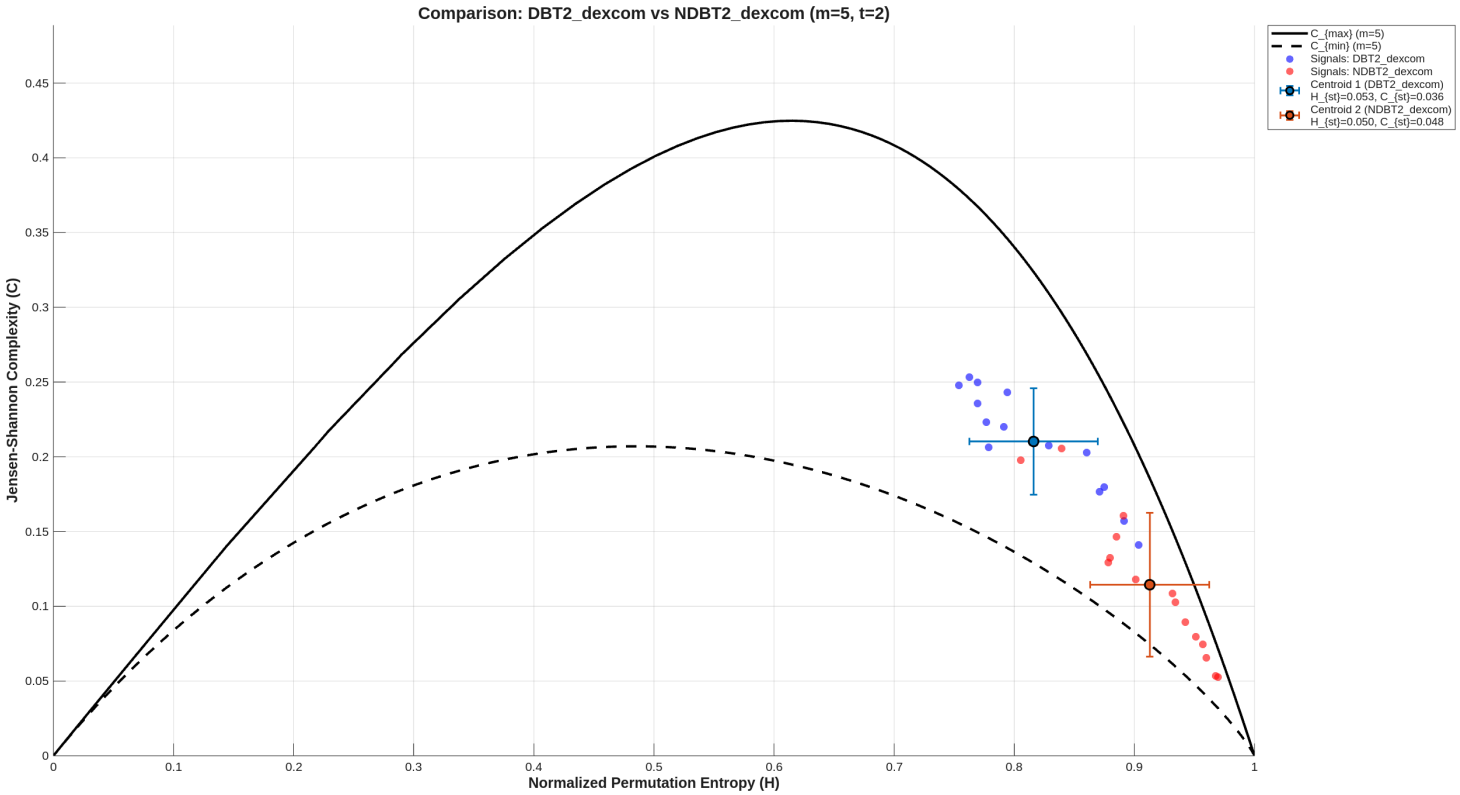


Figure 14: HxC feature plan of the centroids of the CGMacros dataset with dexcom probe

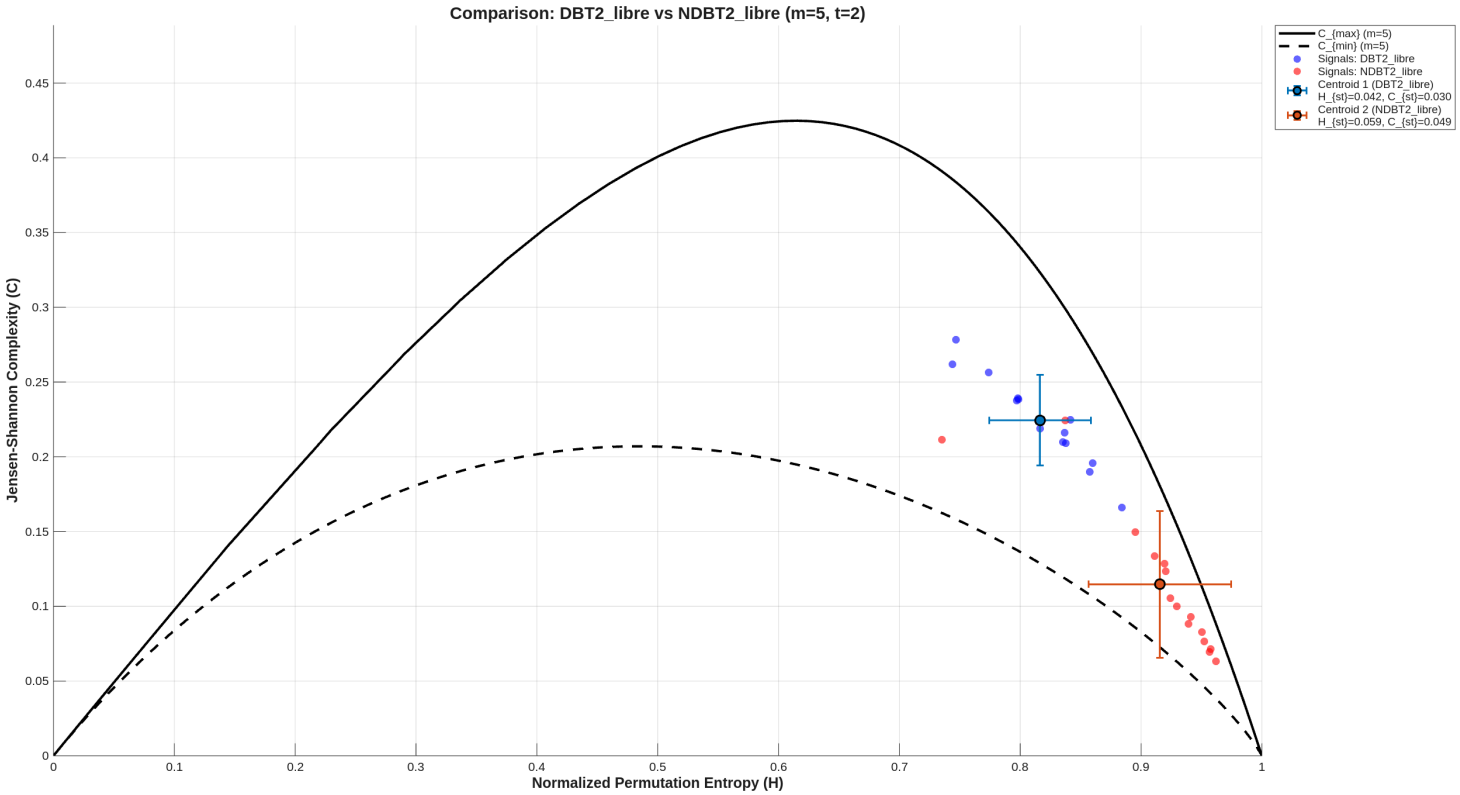


Figure 15: HxC feature plan of the centroids of the CGMacros dataset with libre probe

As we still observe a good separation bet ween the two classes, we got a bigger variability within each group which may disturb the classification:

Group of patient+probe	**Shannon entropy (mean ± std)**	**Complexity (mean ± std)**
Diabetic + Dexcom	0.8161 ± 0.0533	0.2103 ± 0.0357
Non-diabetic + Dexcom	0.9129 ± 0.0495	0.1144 ± 0.0485
Diabetic + Libre	0.8164 ± 0.0421	0.2244 ± 0.0302
Non-diabetic + Libre	0.9155 ± 0.0591	0.1147 ± 0.0491

Table 7: Mean and standard deviation of the Shannon entropy and complexity for each group of patient depending of the probde

With the regular Kmeans algorithm we obtain:

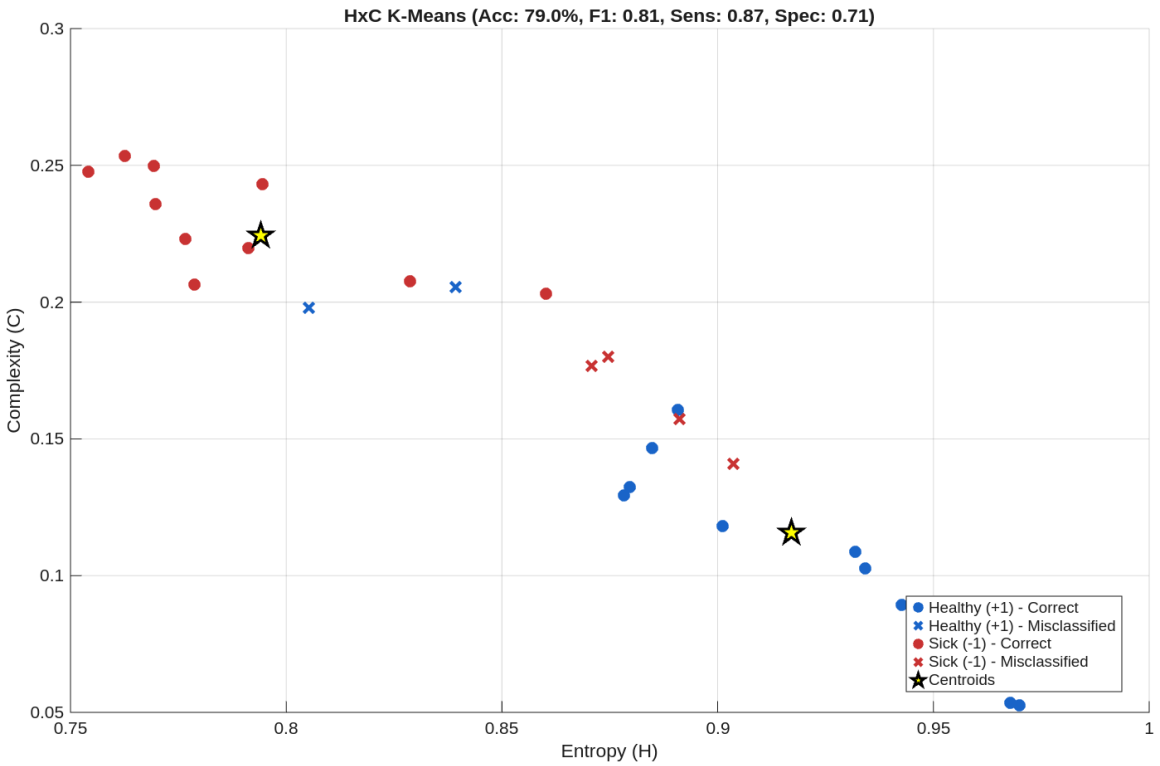


Figure 16: Kmeans algorithm applied on the HxC feature plane of the CGMacros dataset with dexcom probe

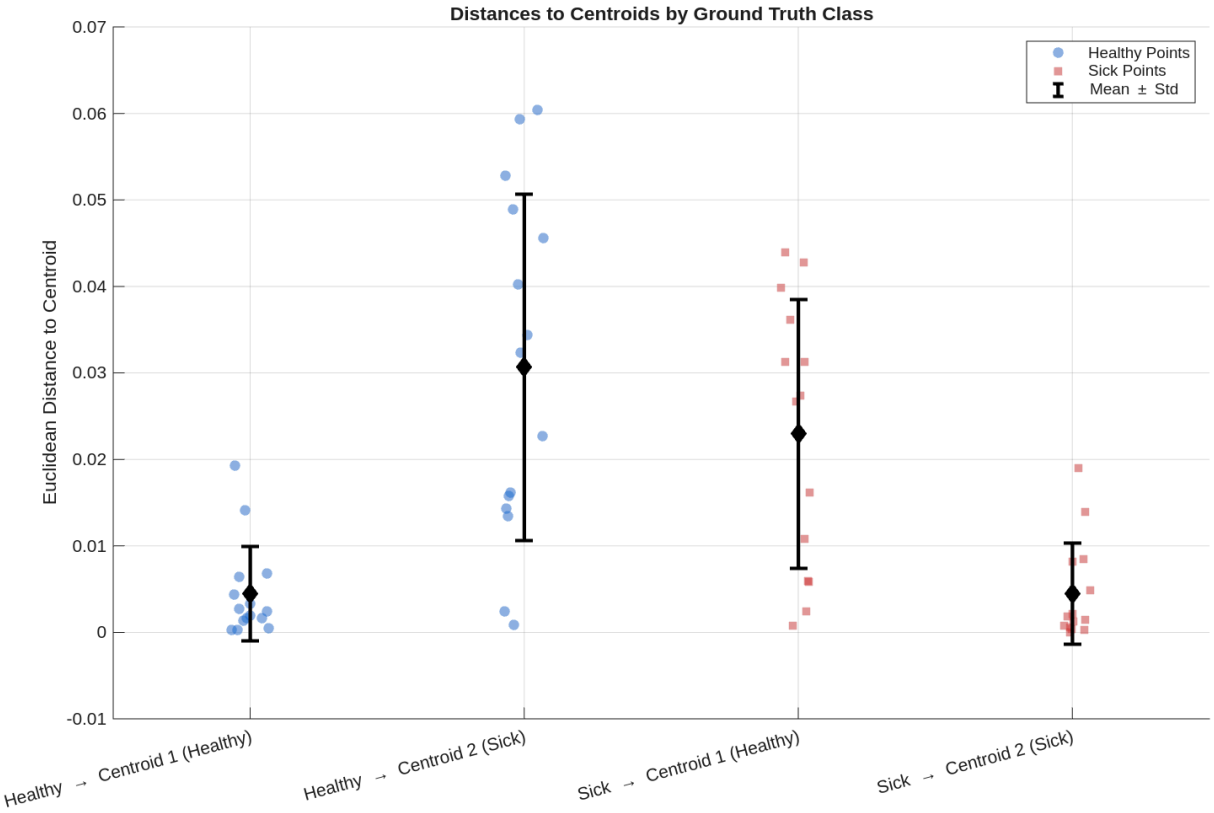


Figure 17: Distance to cluster centroids of the Kmeans algorithm applied on the HxC feature plane of the CGMacros dataset with dexcom probe

Results are much better with the libre probe:

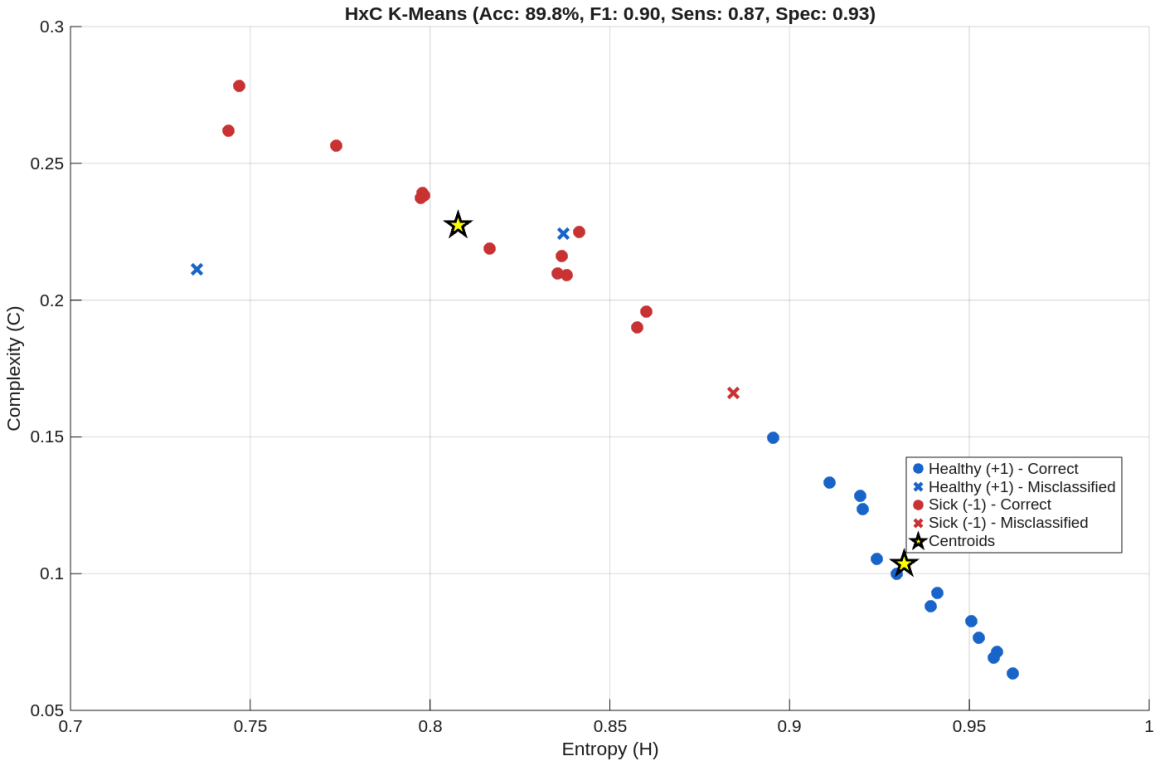


Figure 18: Kmeans algorithm applied on the HxC feature plane of the CGMacros dataset with libre probe

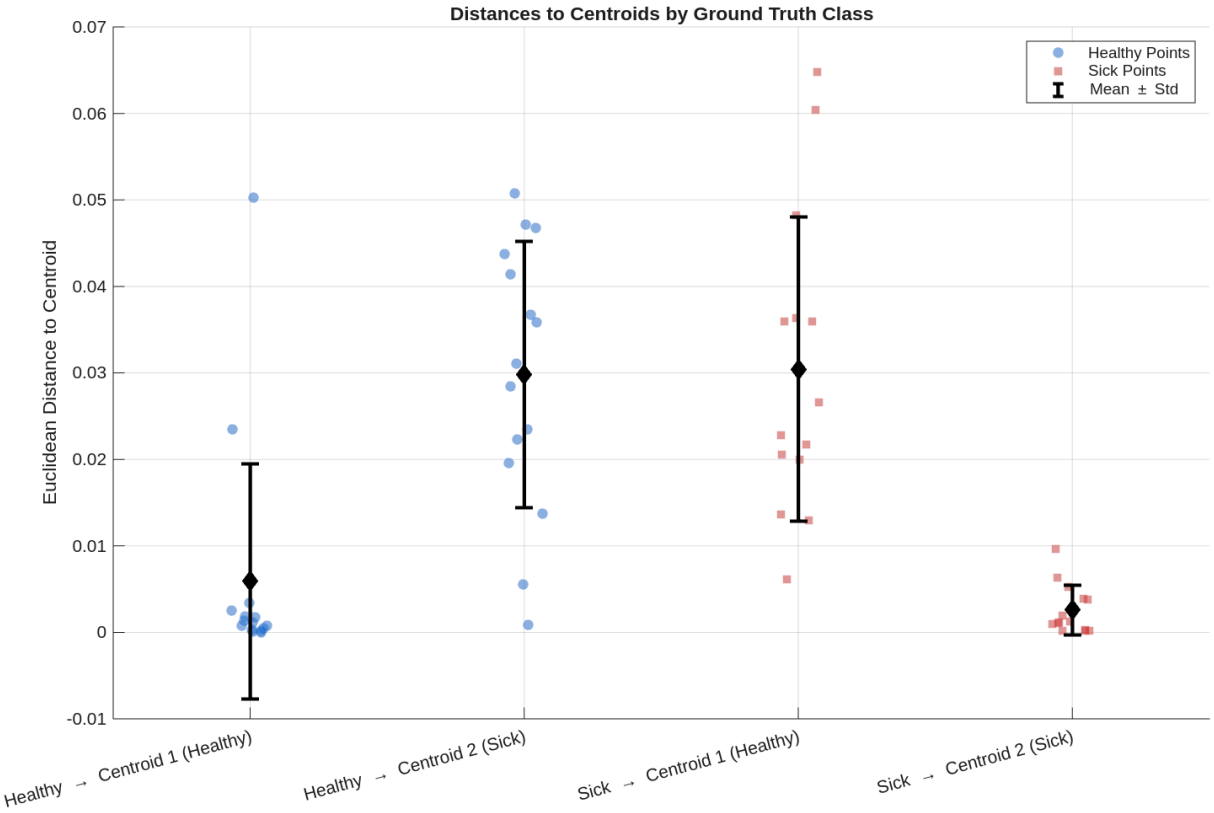


Figure 19: Distance to cluster centroids of the Kmeans algorithm applied on the HxC feature plane of the CGMacros dataset with libre probe

Prob	Accuracy	F1-score	Sensibility	Specificity
Dexcom	0.79	0.81	0.87	0.71
Libre	0.90	0.90	0.87	0.93
Average quality	**0.845**	**0.855**	**0.87**	**0.82**

Table 8: Classification performance of the Kmeans algorithm applied on the HxC feature plane of the CGMacros dataset with dexcom and libre probes

Result : Despite a greater spread in the cloud points, the Kmeans algorithm using Shannon entropy with **(m=5, τ=2)** parameters is still able to find a good separation between the two classes of patients. The classification performance is even better with the libre probe than with the dexcom probe.